# Data Science

## COMP5122M

Data Quality

Duygu Sarikaya

School of Computing, University of Leeds

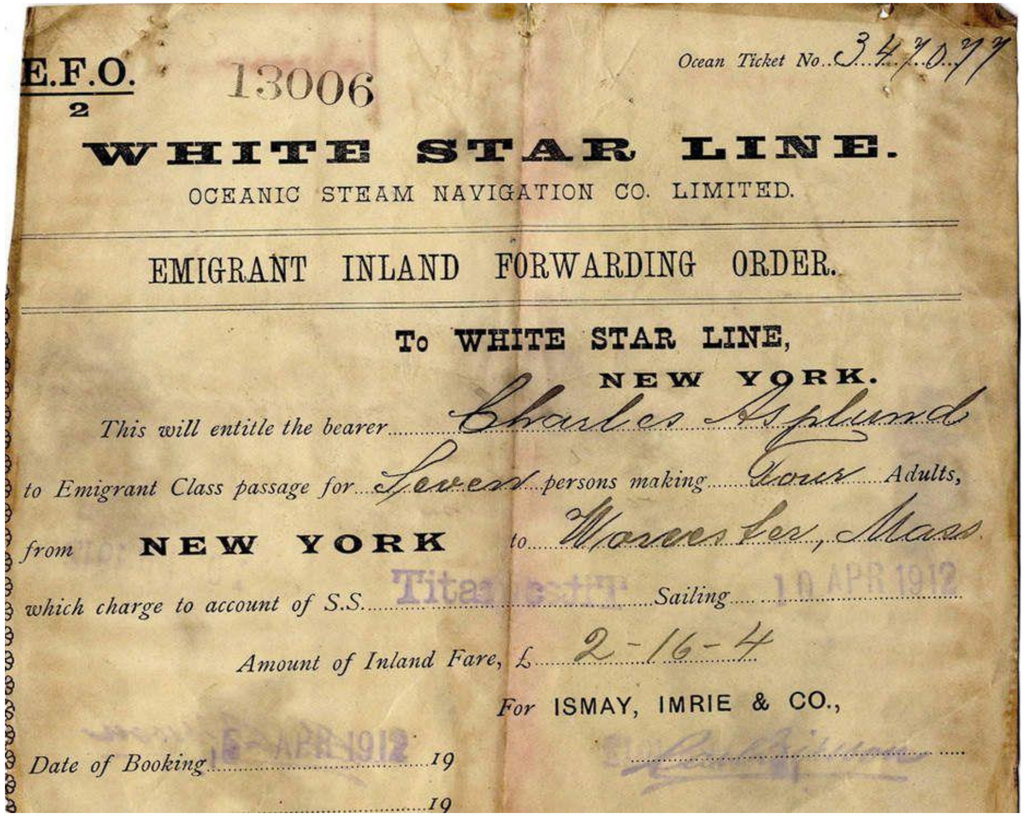
# What is Data?

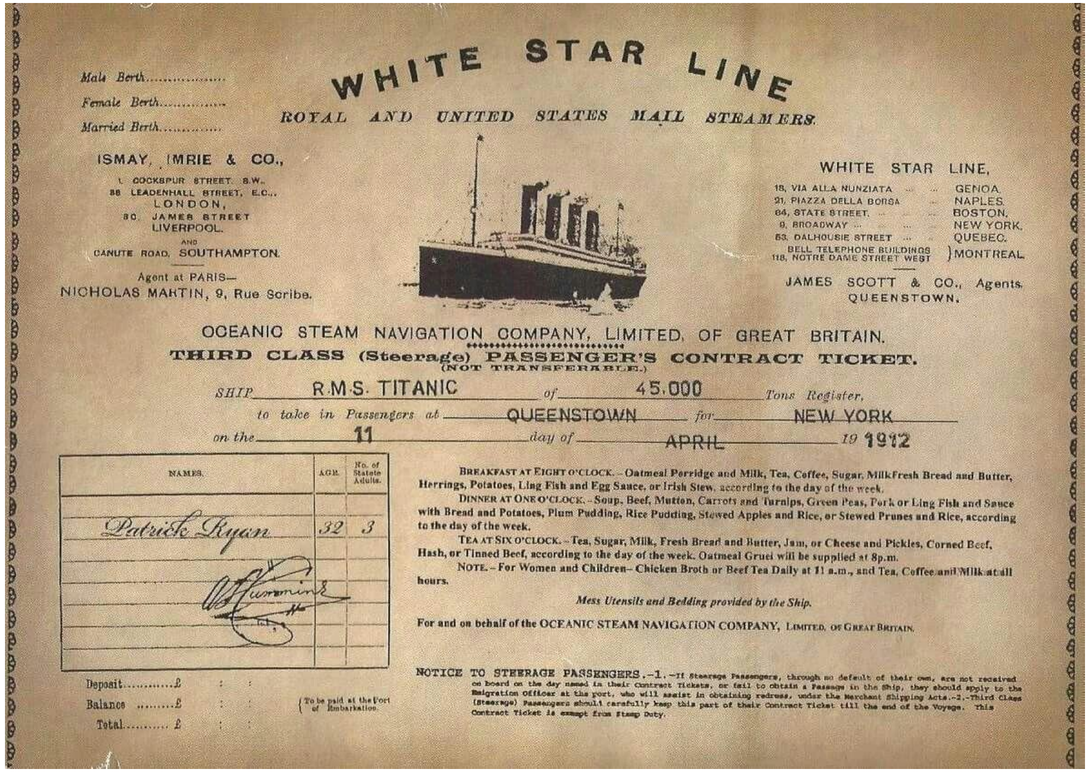
Objective: Predict the survival of the passengers aboard RMS Titanic.

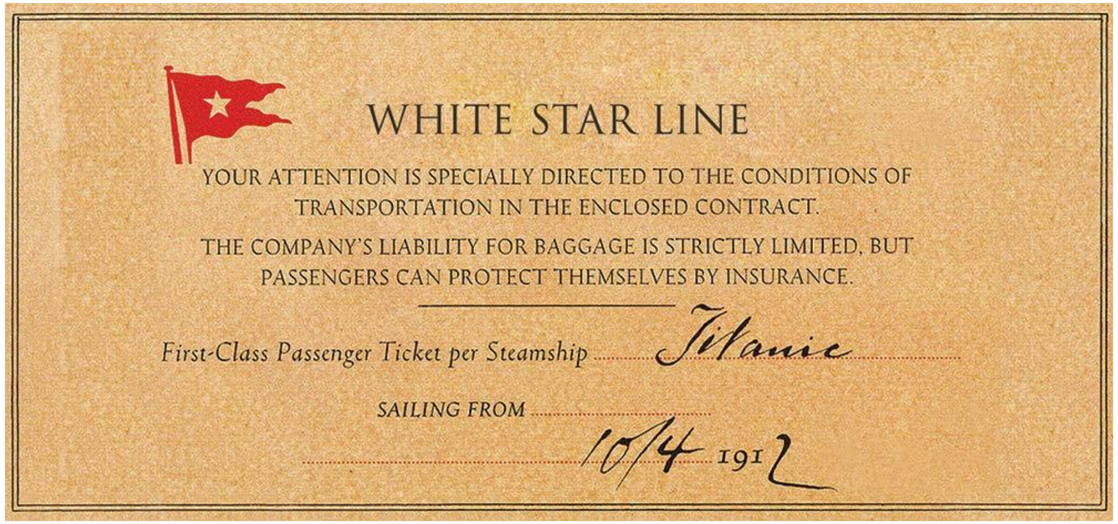
Dataset: Titanic DataSet

- provides the data on all the passengers who were aboard the RMS Titanic when it sank on 15 April 1912 after colliding with an iceberg in the North Atlantic ocean.
    - a combination of variables based on personal characteristics such as age, class of ticket and gender, and tests one's classification skills.
    - 891 rows and 12 columns

# What is Data?


E.F.O. 2
13006
Ocean Ticket No. 347077
WHITE STAR LINE.
OCEANIC STEAM NAVIGATION CO. LIMITED.
EMIGRANT INLAND FORWARDING ORDER.
To WHITE STAR LINE,
NEW YORK.
This will entitle the bearer Charles Asplund
to Emigrant Class passage for Seven persons making Four Adults,
from NEW YORK to Worcester, Mass
which charge to account of S.S. Titanic Sailing 10 APR 1912
Amount of Inland Fare, £ 2-16-4
For ISMAY, IMRIE & CO.,
Date of Booking 6 APR 1912 19
19


Male Berth
Female Berth
Married Berth
WHITE STAR LINE
ROYAL AND UNITED STATES MAIL STEAMERS.
ISMAY, IMRIE & CO.,
1, COCKSPUR STREET, S.W.,
38, LEADENHALL STREET, E.C.,
LONDON,
30, JAMES STREET
LIVERPOOL.
AND
CANUTE ROAD, SOUTHAMPTON.
Agent at PARIS—
NICHOLAS MARTIN, 9, Rue Scribe.
WHITE STAR LINE,
18, VIA ALLA NUNZIATA ... GENOA.
21, PIAZZA DELLA BORSA ... NAPLES.
84, STATE STREET ... BOSTON.
9, BROADWAY ... NEW YORK.
53, DALHOUSIE STREET ... QUEBEC.
BELL TELEPHONE BUILDINGS 118, NOTRE DAME STREET WEST } MONTREAL.
JAMES SCOTT & CO., Agents, QUEENSTOWN.
OCEANIC STEAM NAVIGATION COMPANY, LIMITED, OF GREAT BRITAIN.
THIRD CLASS (Steerage) PASSENGER'S CONTRACT TICKET.
(NOT TRANSFERABLE.)
SHIP R.M.S. TITANIC of 45,000 Tons Register,
to take in Passengers at QUEENSTOWN for NEW YORK
on the 11 day of APRIL 19 1912
NAMES. AGE. No. of Statute Adults.
Patrick Ryan 32 3
Deposit £
Balance £
Total £
To be paid at the Port of Embarkation.
BREAKFAST AT EIGHT O'CLOCK.—Oatmeal Porridge and Milk, Tea, Coffee, Sugar, Milk Fresh Bread and Butter, Herrings, Potatoes, Ling Fish and Egg Sauce, or Irish Stew, according to the day of the week.
DINNER AT ONE O'CLOCK.—Soup, Beef, Mutton, Carrots and Turnips, Green Peas, Pork or Ling Fish and Sauce with Bread and Potatoes, Plum Pudding, Rice Pudding, Stewed Apples and Rice, or Stewed Prunes and Rice, according to the day of the week.
TEA AT SIX O'CLOCK.—Tea, Sugar, Milk, Fresh Bread and Butter, Jam, or Cheese and Pickles, Corned Beef, Hash, or Tinned Beef, according to the day of the week. Oatmeal Gruel will be supplied at 8p.m.
NOTE.—For Women and Children—Chicken Broth or Beef Tea Daily at 11 a.m., and Tea, Coffee and Milk at all hours.
Mess Utensils and Bedding provided by the Ship.
For and on behalf of the OCEANIC STEAM NAVIGATION COMPANY, LIMITED, OF GREAT BRITAIN.
NOTICE TO STEERAGE PASSENGERS.—1.—If Steerage Passengers, through no default of their own, are not received on board on the day named in their Contract Tickets, or fail to obtain a Passage in the Ship, they should apply to the Emigration Officer at the port, who will assist in obtaining redress, under the Merchant Shipping Acts.—2.—Third Class (Steerage) Passengers should carefully keep this part of their Contract Ticket till the end of the Voyage. This Contract Ticket is exempt from Stamp Duty.


WHITE STAR LINE
YOUR ATTENTION IS SPECIALLY DIRECTED TO THE CONDITIONS OF TRANSPORTATION IN THE ENCLOSED CONTRACT.
THE COMPANY'S LIABILITY FOR BAGGAGE IS STRICTLY LIMITED, BUT PASSENGERS CAN PROTECT THEMSELVES BY INSURANCE.
First-Class Passenger Ticket per Steamship Titanic
SAILING FROM
10/4 1912

# What is Data?

"A datum is a single measurement of something on a scale that is understandable to both the recorder and the reader. Data are multiple such measurements."

# Where does it come from?

- **Internal sources:** already collected by or is part of the overall data collection of your organization.
  For example: business-centric data that is available in the organization data base to record day to day operations; scientific or experimental data.
- **Existing External Sources:** available in ready to read format from an outside source for free or for a fee.
  For example: public government databases, stock market data, Yelp reviews, [your favorite sport]-reference.
- **External Sources Requiring Collection Efforts**: available from external source but acquisition requires special processing.
  For example: data appearing only in print form, or data on websites.

# Where does it come from?

How to get data generated, published or hosted online:

- **API (Application Programming Interface):** using a prebuilt set of functions developed by a company to access their services. Often pay to use. For example: Google Map API, Facebook API, Twitter API
- **RSS (Rich Site Summary):** summarizes frequently updated online content in standard format. Free to read if the site has one. For example: news-related sites, blogs
- **Web scraping:** using software, scripts or by-hand extracting data from what is displayed on a page or what is contained in the HTML file (often in tables).

# Web Scraping

- **Why do it?** Older government or smaller news sites might not have APIs for accessing data, or publish RSS feeds or have databases for download. Or, you don’t want to pay to use the API or the database.
- **How do you do it?**
- **Should you do it?**
  - You just want to explore: Are you violating their terms of service? Privacy concerns for website and their clients?
  - You want to publish your analysis or product: Do they have an API or fee that you are bypassing? Are they willing to share this data? Are you violating their terms of service? Are there privacy concerns?

# What type of data?

## Key Data Properties

Structure -- the “shape” of a data file

Granularity -- how fine/coarse is each datum

Scope -- how (in)complete is the data

Temporality -- how is the data situated in time

Faithfulness -- how well does the data capture “reality”

# What type of data?

## Key Data Properties

### Structure -- the “shape” of a data file

e.g. Tab separated values

A common table file format.

- Records are delimited by a newline: `'\n'`, `"\r\n"`
- Fields are delimited by `'\t'` (tab)

[`pd.read_csv`](pd.read_csv): Need to specify

```
delimiter='\t'
```

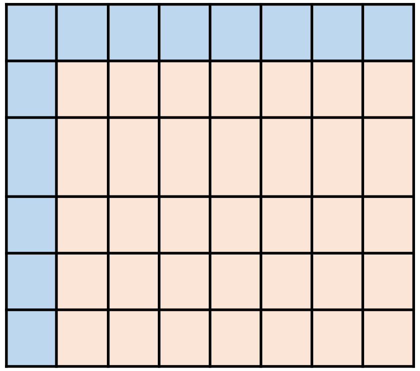
# What type of data?

## Key Data Properties

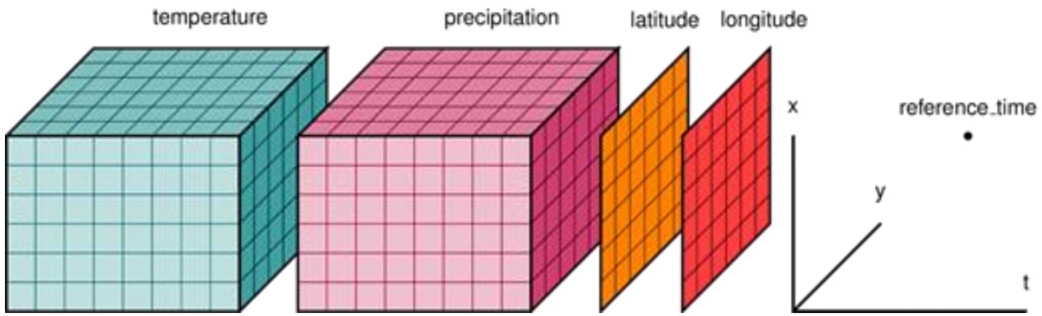
**Structure** -- the "shape" of a data file

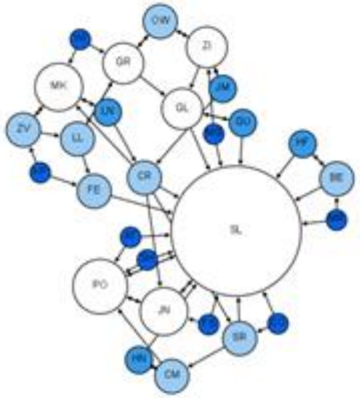
Data come in many different shapes.

Rectangular data



Non-rectangular data



# How is the data stored?

How is your data represented and stored (data format)?

- **Tabular Data:** a dataset that is a two-dimensional table, where each row typically represents a single data record, and each column represents one type of measurement (csv, dat, xlsx, etc.).
- **Structured Data:** each data record is presented in a form of a [possibly complex and multi-tiered] dictionary (json, xml, etc.)
- **Semistructured Data:** not all records are represented by the same set of keys or some data records are not represented using the key-value pair structure.

# What type of data? How is it stored?

What kind of values are in your data (data types)?

Simple or atomic:

- **Numeric:** integers, floats
- **Boolean:** binary or true false values
- **Strings:** sequence of symbols

# What type of data? How is it stored?

What kind of values are in your data (data types)? Compound, composed of a bunch of atomic types:

- **Date and time:** compound value with a specific structure
- **Lists:** a list is a sequence of values
- **Dictionaries:** A dictionary is a collection of key-value pairs, a pair of values *x* : *y* where *x* is usually a string called the key representing the "name" of the entry, and *y* is a value of any type.

Example: Student record: what are *x* and *y*?

- First: Kevin
- Last: Rader
- Classes: [CS-109A, STAT139]

CDC Morbidity and Mortality Weekly Report (MMWR) 03/25/2022.

## Summary

**What is already known about this topic?**

The number of reported U.S. tuberculosis (TB) cases decreased sharply in 2020, possibly related to multiple factors associated with the COVID-19 pandemic.

**What is added by this report?**

Reported TB incidence (cases per 100,000 persons) increased 9.4%, from 2.2 during 2020 to 2.4 during 2021 but was lower than incidence during 2019 (2.7). Increases occurred among both U.S.-born and non–U.S.-born persons.

**What are the implications for public health practice?**

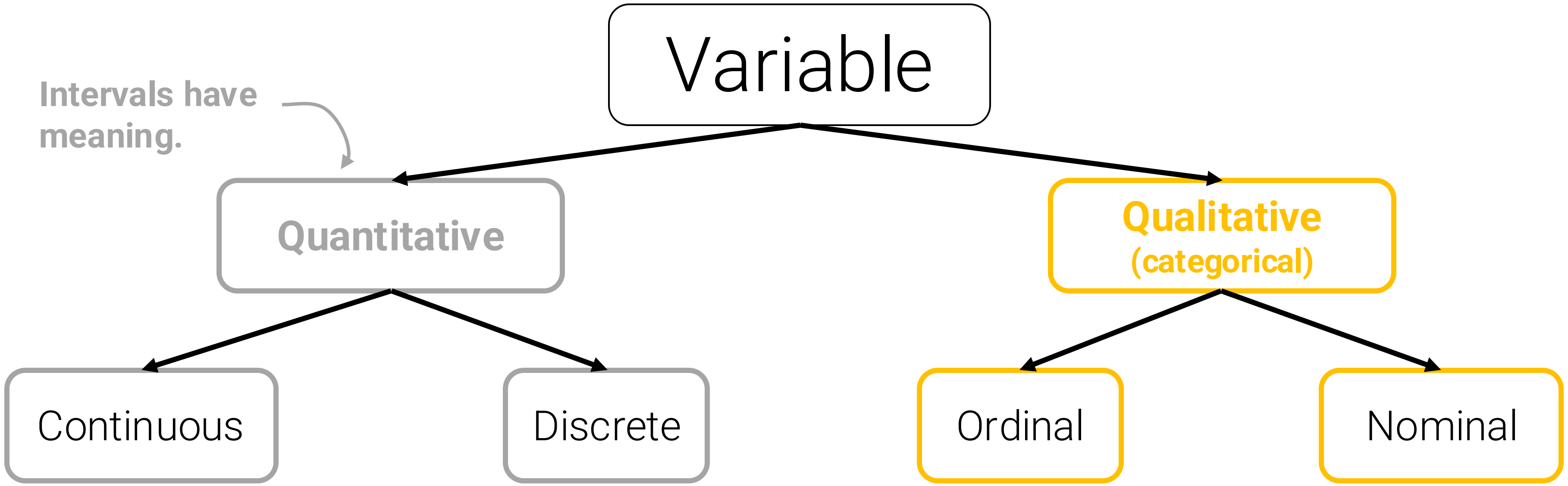
Factors contributing to changes in reported TB during 2020–2021 likely include an actual reduction in TB incidence as well as delayed or missed TB diagnoses. Timely evaluation and treatment of TB and latent tuberculosis infection remain critical to achieving U.S. TB elimination.

CDC: **source**, 2021

What is **incidence**?

The number of "cases per 100,000 persons using mid-year population estimates from the U.S. Census Bureau."

# What type of data?




**Continuous:** Could be measured to arbitrary precision.

**Examples:**
- Price
- Temperature

**Discrete:** Finite possible values

**Examples:**
- Number of siblings
- Yrs of education

**Ordinal:** Categories w/ordered levels; no consistent meaning to difference

**Examples:**
- Preferences
- Level of education

**Nominal:** Categories w/ no specific ordering.

**Examples:**
- Political Affiliation
- Student ID

# What type of data?

Many variables do not sit neatly in one of these categories!!

Variable

Intervals have meaning.

Quantitative

Qualitative (categorical)

Continuous

Could be measured to arbitrary precision.

**Examples:**
- Price
- Temperature

Discrete

Finite possible values

**Examples:**
- Number of siblings
- Yrs of education

Ordinal

Categories w/ordered levels; no consistent meaning to difference

**Examples:**
- Preferences
- Level of education

Nominal

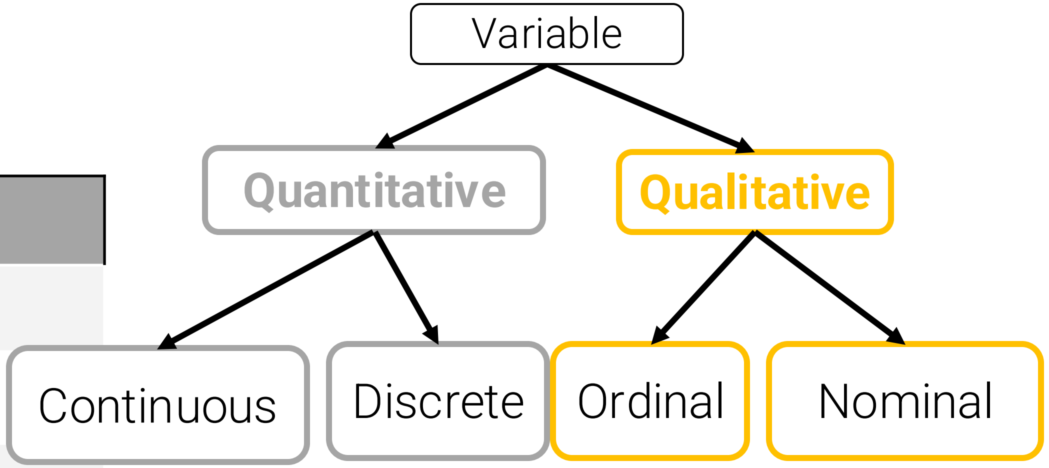
Categories w/ no specific ordering.

**Examples:**
- Political Affiliation
- Cal ID number

# What type of data?

What is the feature type of each variable?

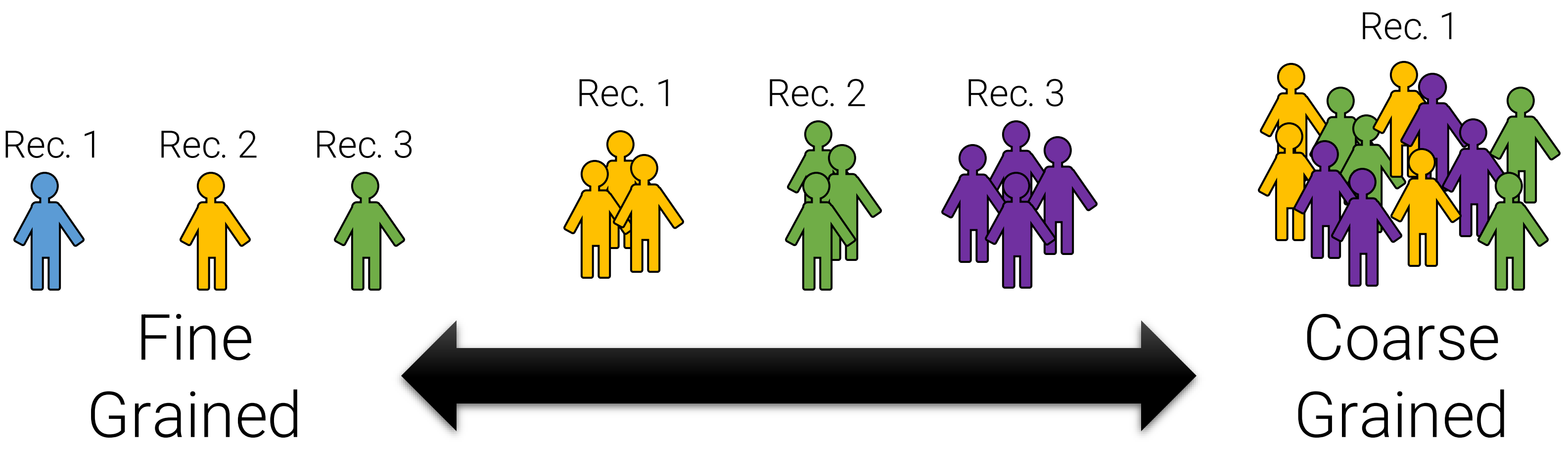
| Q | Variable | Feature Type |
|---|---|---|
| 1 | $CO_2$ level (ppm) | |
| 2 | Number of siblings | |
| 3 | GPA | |
| 4 | Income bracket (low, med, high) | |
| 5 | Race/Ethnicity | |
| 6 | Shirt Size (S,M,L) | |
| 7 | Yelp Star Rating | |



# What type of data?

## Key Data Properties

Granularity -- how fine/coarse is each datum



# What type of data?

## Key Data Properties

## Scope -- how (in)complete is the data

Will my data be enough to answer my question?

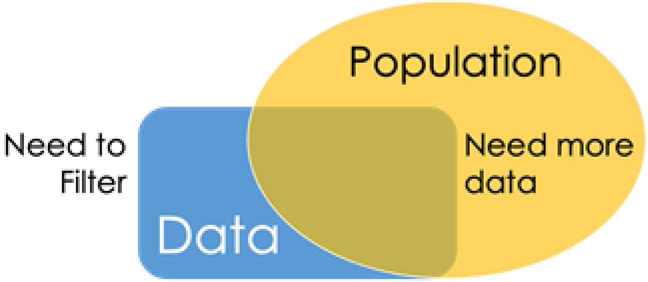
- **Example**: I am interested in studying crime in California but I only have Berkeley crime data.
- **Solution**: collect more data/change research question

Is my data too broad?

- **Example**: I am interested in student grades for Algorithms but have student grades for all Algorithms classes (grad + undergrad).
- **Solution: Filtering** ⇒ Implications on sample?
    - If the data is a sample I may have poor coverage after filtering (More on this next week)

Does my data cover the right time frame?

- Which brings us to **Temporality**




# What type of data?

Temporality -- how is the data situated in time

**Data changes** – when was the data collected/last updated?

**Periodicity** — Is there periodicity? Diurnal (24-hr) patterns?

What is the meaning of the time and date fields? A few options:

- When the “event” happened?
- When the data was collected or was entered into the system?
- Date the data was copied into a database? (look for many matching timestamps)

# What type of data?

**Faithfulness** -- how well does the data capture “reality”

Does my data contain **unrealistic or “incorrect” values**?

- Dates in the future for events in the past
- Locations that don’t exist
- Negative counts
- Misspellings of names
- Large outliers

Does my data violate **obvious dependencies**?

- E.g., age and birthday don’t match

Was the data **entered by hand**?

- Spelling errors, fields shifted …
- Did the form require all fields or provide default values?

Are there obvious signs of **data falsification**?

- Repeated names, fake looking email addresses, repeated use of uncommon names or fields.

# What type of data?

**Faithfulness** -- how well does the data capture "reality"

**Truncated data**
Early Microsoft Excel limits: 65536 Rows, 255 Columns

**Duplicated Records or Fields**
Identify and eliminate (use primary key).

- Be aware of consequences in analysis when using data with inconsistencies.
- Understand the potential implications for how data were collected.

**Spelling Errors**
Apply corrections or drop records not in a dictionary

**Units not specified or consistent**
Infer units, check values are in reasonable ranges for data

**Time Zone Inconsistencies**
Convert to a common timezone (e.g., UTC)

**Missing Data???**

Examples

| | |
|---|---|
| " " | 1970, 2000 |
| 0, -1 | NaN |
| 999, 12345 | Null |

NaN: "Not a Number"

# What type of data?

**A. Drop records** with missing values

- Probably most common
- **Caution:** check for biases induced by dropped values
  - Missing or corrupt records might be related to something of interest

**B. Keep as** NaN

**C. Imputation/Interpolation**: Inferring missing values

- **Average Imputation:** replace with an average value
  - Which average? Often use closest related subgroup mean.
- **Hot deck imputation:** replace with a random value
- **Regression imputation:** replace with a predicted value, using some model
- **Multiple imputation:** replace with multiple random values.

# What type of data?

Choice affects bias and uncertainty quantification (large statistics literature)

**Essential question:** why are the records missing?

**A. Drop records** with missing values

- Probably most common
- **Caution:** check for biases induced by dropped values
  - Missing or corrupt records might be related to something of interest

**B. Keep as** NaN

**C. Imputation/Interpolation**: Inferring missing values

- **Average Imputation:** replace with an average value
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# Common Issues

A few good generic questions to ask are as follows:

- How big is the dataset?
- Is this the entire dataset?
- Is this data representative enough? For example, maybe data was only collected for a subset of users.
- Are there likely to be gross outliers or extraordinary sources of noise? For example, 99% of the traffic from a web server might be a single denial-of-service attack.
- Might there be artificial data inserted into the dataset? This happens a lot in industrial settings.
- Are there any fields that are unique identifiers? These are the fields you might use for joining between datasets, etc.
- Are the supposedly unique identifiers actually unique? What does it mean if they aren't?
- If there are two datasets A and B that need to be joined, what does it mean if something in A doesn't matching anything in B?
- When data entries are blank, where does that come from?
- How common are blank entries?

# Common Issues

Common issues with data:

- Missing values: how do we fill in?
- Wrong values: how can we detect and correct?
- Messy format
- Not usable: the data cannot answer the question posed

# Common Issues

Common causes of messy data are:

- Column headers are values, not variable names
- Variables are stored in both rows and columns
- Multiple variables are stored in one column/entry
- Multiple types of experimental units stored in same table

In general, we want each file to correspond to a dataset, each column to represent a single variable and each row to represent a single observation.

We want to **tabularize** the data.

# Common Issues

Sometimes your data comes in multiple files:

- Often data will reference other pieces of data.
- Alternatively, you will collect multiple pieces of related data.

Use `pd.merge` to join data on **keys**.

**Primary key**: the *column* or *set of columns* in a table that *uniquely* determine the values in the remaining columns

- Primary keys are unique
- Examples: SSN, ProductIDs, …

**Foreign keys**: the *column* or *set of columns* that reference primary keys in other tables.

Primary Key

Customers.csv

| CustID | Addr |
|---|---|
| 171345 | Harmon.. |
| 281139 | Main .. |

Primary Key

Orders.csv

| OrderNum | CustID | Date |
|---|---|---|
| 1 | 171345 | 8/21/2017 |
| 2 | 281139 | 8/30/2017 |

Products.csv

| ProdID | Cost |
|---|---|
| 42 | 3.14 |
| 999 | 2.72 |

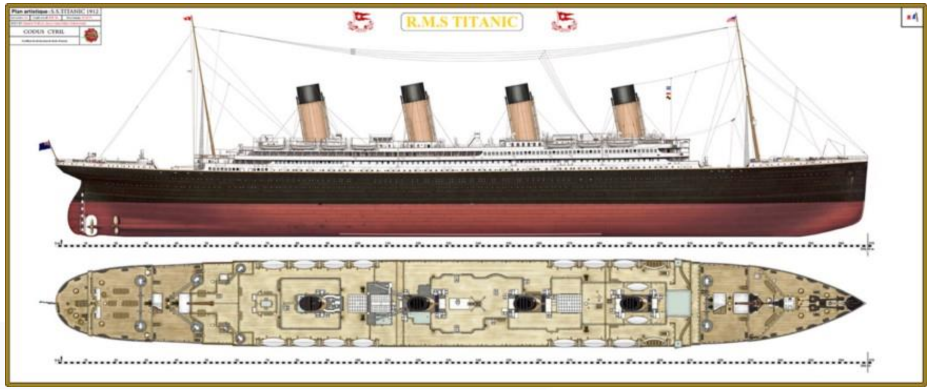
Primary Key

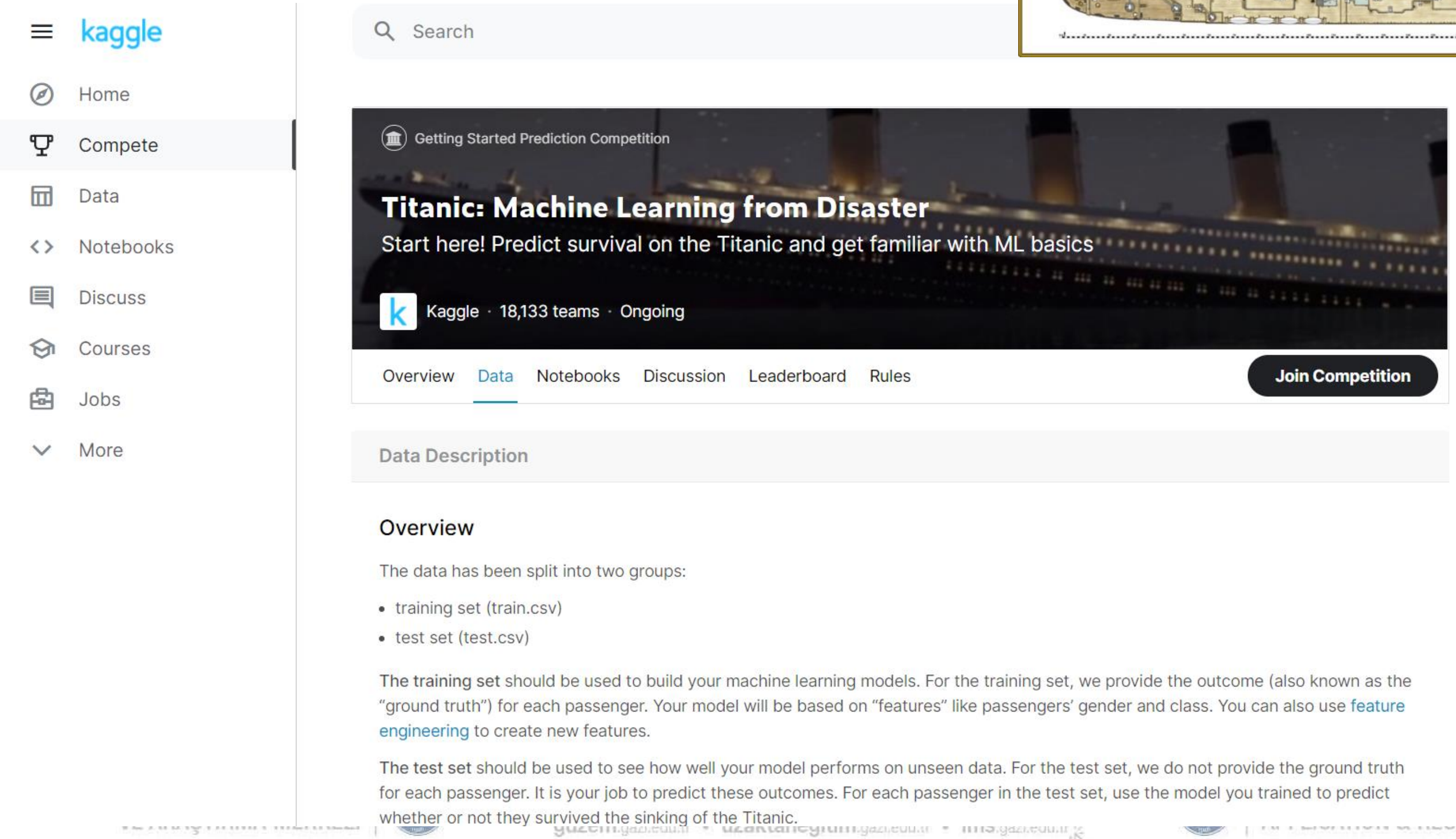
Purchases.csv

| OrderNum | ProdID | Quantity |
|---|---|---|
| 1 | 42 | 3 |
| 1 | 999 | 2 |
| 2 | 42 | 1 |

# Where to find Datasets?

https://www.kaggle.com/c/titanic/data



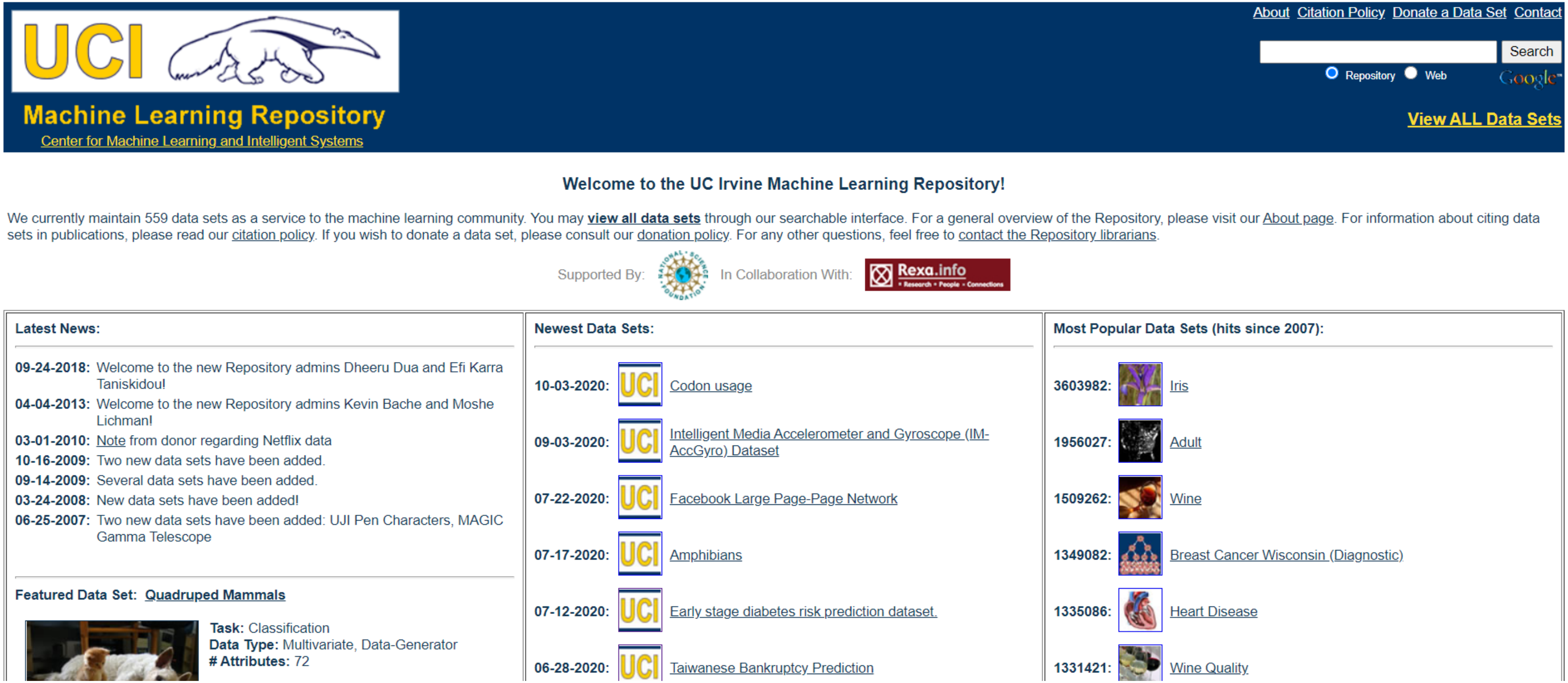



# Where to find Datasets?

| PassengerId | Survived | Pclass | Name | Sex | Age | SibSp | Parch | Ticket | Fare | Cabin | Embarked |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 3 | Braund, Mr. Owen Harris | male | 22 | 1 | 0 | A/5 21171 | 7.25 | | S |
| 2 | 1 | 1 | Cumings, Mrs. John Bradley (Florence Briggs Thayer) | female | 38 | 1 | 0 | PC 17599 | 71.2833 | C85 | C |
| 3 | 1 | 3 | Heikkinen, Miss. Laina | female | 26 | 0 | 0 | STON/O2. 3101282 | 7.925 | | S |
| 4 | 1 | 1 | Futrelle, Mrs. Jacques Heath (Lily May Peel) | female | 35 | 1 | 0 | 113803 | 53.1 | C123 | S |
| 5 | 0 | 3 | Allen, Mr. William Henry | male | 35 | 0 | 0 | 373450 | 8.05 | | S |
| 6 | 0 | 3 | Moran, Mr. James | male | | 0 | 0 | 330877 | 8.4583 | | Q |
| 7 | 0 | 1 | McCarthy, Mr. Timothy J | male | 54 | 0 | 0 | 17463 | 51.8625 | E46 | S |
| 8 | 0 | 3 | Palsson, Master. Gosta Leonard | male | 2 | 3 | 1 | 349909 | 21.075 | | S |
| 9 | 1 | 3 | Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg) | female | 27 | 0 | 2 | 347742 | 11.1333 | | S |
| 10 | 1 | 2 | Nasser, Mrs. Nicholas (Adele Achem) | female | 14 | 1 | 0 | 237736 | 30.0708 | | C |
| 11 | 1 | 3 | Sandstrom, Miss. Marguerite Rut | female | 4 | 1 | 1 | PP 9549 | 16.7 | G6 | S |
| 12 | 1 | 1 | Bonnell, Miss. Elizabeth | female | 58 | 0 | 0 | 113783 | 26.55 | C103 | S |
| 13 | 0 | 3 | Saundercock, Mr. William Henry | male | 20 | 0 | 0 | A/5. 2151 | 8.05 | | S |
| 14 | 0 | 3 | Andersson, Mr. Anders Johan | male | 39 | 1 | 5 | 347082 | 31.275 | | S |
| 15 | 0 | 3 | Vestrom, Miss. Hulda Amanda Adolfina | female | 14 | 0 | 0 | 350406 | 7.8542 | | S |
| 16 | 1 | 2 | Hewlett, Mrs. (Mary D Kingcome) | female | 55 | 0 | 0 | 248706 | 16 | | S |
| 17 | 0 | 3 | Rice, Master. Eugene | male | 2 | 4 | 1 | 382652 | 29.125 | | Q |
| 18 | 1 | 2 | Williams, Mr. Charles Eugene | male | | 0 | 0 | 244373 | 13 | | S |
| 19 | 0 | 3 | Vander Planke, Mrs. Julius (Emelia Maria Vandemoortele) | female | 31 | 1 | 0 | 345763 | 18 | | S |
| 20 | 1 | 3 | Masselmani, Mrs. Fatima | female | | 0 | 0 | 2649 | 7.225 | | C |

# Where to find Datasets?

UCI Machine Learning Repository - 559 data sets https://archive.ics.uci.edu/ml/index.php



# Where to find Datasets?

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Search

Repository Web Google

UCI

Machine Learning Repository

Center for Machine Learning and Intelligent Systems

View ALL Data Sets

## Early stage diabetes risk prediction dataset. Data Set

*Download*: Data Folder, Data Set Description

**Abstract**: This dataset contains the sign and symptpom data of newly diabetic or would be diabetic patient.

| | | | | | |
|---|---|---|---|---|---|
| **Data Set Characteristics:** | Multivariate | **Number of Instances:** | 520 | **Area:** | Computer |
| **Attribute Characteristics:** | N/A | **Number of Attributes:** | 17 | **Date Donated** | 2020-07-12 |
| **Associated Tasks:** | Classification | **Missing Values?** | Yes | **Number of Web Hits:** | 17904 |

**Source:**

1. M M Faniqul Islam,
2. Rahatara Ferdousi,
3.Sadikur Rahman,
and Humayra
4.Yasmin Bushra

1. Queen Mary University of London, United Kingdom
m.islam '@' smd17.qmul.ac.uk
2. Metropolitan University Sylhet, Bangladesh
rahatara '@' metrouni.edu.bd
3 . Metropolitan University Sylhet, Bangladesh
rahmansadik004 '@' gmail.com
4 .Metropolitan University Sylhet, Bangladesh
humayrabushra234 '@' gmail.com

# Where to find Datasets?

### Data Set Information:

This has been col-
lected using direct questionnaires from the patients of Sylhet Diabetes Hospital in Sylhet, Bangladesh and approved by a doctor.

### Attribute Information:

Age 1.20-65
Sex 1. Male, 2.Female
Polyuria 1.Yes, 2.No.
Polydipsia 1.Yes, 2.No.
sudden weight loss 1.Yes, 2.No.
weakness 1.Yes, 2.No.
Polyphagia 1.Yes, 2.No.
Genital thrush 1.Yes, 2.No.
visual blurring 1.Yes, 2.No.
Itching 1.Yes, 2.No.
Irritability 1.Yes, 2.No.
delayed healing 1.Yes, 2.No.
partial paresis 1.Yes, 2.No.
muscle sti□ness 1.Yes, 2.No.
Alopecia 1.Yes, 2.No.
Obesity 1.Yes, 2.No.
Class 1.Positive, 2.Negative.

### Relevant Papers:

Likelihood Prediction of Diabetes at Early Stage Using Data Mining Techniques
[Web Link]
Authors and affiliations
M. M. Faniqul IslamEmail
Rahatara Ferdousi
Sadikur Rahman
Humayra Yasmin Bushra

# Where else to find Datasets?

Academic Torrents
r/datasets
datahub.io
aws.amazon.com/datasets
Kaggle Data Sources
Kaggle Datasets
World Bank Data
NYC Taxi data
Open Data Philly Connecting people with data for Philadelphia
National Climatic Data Center - NOAA
ClimateData.us
UNICEF Data
undata
NASA SocioEconomic Data and Applications Center - SEDAC
San Fransisco Government Open Data

# References:

- Pavlos Protopapas, Kevin Rader and Chris Tanner (data)
- Joseph E. Gonzalez
- Narges Norouzi